Math140C - HW #3

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Q23

$f(0,1,-1) = 0+1-1 = 0$. Next,

$$[f'(x,y_1,y_2)] = [2xy + e^x, x^2, 1],$$

so $[D_1 f(0,1,-1)] = 1$. Namely, in the language of the implicit function theorem, A_x is invertible. Hence the implicit function theorem applies; there is a neighborhood $W \subset \mathbb{R}^2$ of $(1,-1)$ and function $g : W \to \mathbb{R}$ such that $g(1,-1) = 0$ and

$$f(g(y_1,y_2),y_1,y_2) = 0, \quad ((y_1,y_2) \in W)$$

Furthermore, the implicit function theorem gives the formula

$$\begin{aligned}[g'(1,-1)] &= -[D_1 f(0,1,-1)]^{-1}[[D_2 f(0,1,-1)],[D_3 f(0,1,-1)]] \\ &= -1[0,1] = [0,-1]\end{aligned}$$

In other words, $[D_1 g(1,-1)] = 0$ and $[D_2 g(1,-1)] = -1$.

Q27

(a)

$$\begin{aligned}\limsup_{(x,y)\to(0,0)} \left|\frac{xy(x^2-y^2)}{x^2+y^2}\right| &\le \lim_{(x,y)\to(0,0)} \left|\frac{x^3 y}{x^2+y^2}\right| + \lim_{(x,y)\to(0,0)} \left|\frac{xy^3}{x^2+y^2}\right| \\ &= \lim_{(x,y)\to(0,0)} |xy| \left|\frac{x^2}{x^2+y^2}\right| + \lim_{(x,y)\to(0,0)} |xy| \left|\frac{y^2}{x^2+y^2}\right| \\ &\le 2 \lim_{(x,y)\to(0,0)} |xy| = 0,\end{aligned}$$

So $\lim_{(x,y)\to(0,0)} f(x,y) = 0$, meaning $f(x,y)$ is continuous at $(0,0)$. Of course, f is continuous away from $(0,0)$ because it is rational. Away from $(0,0)$, one can calculate using the quotient rule

$$[D_1 f(x,y)] = \frac{y(x^4+4x^2y^2-y^4)}{(x^2+y^2)^2} \tag{1}$$

$$[D_2 f(x,y)] = \frac{x^5-4x^3y^2-xy^4}{(x^2+y^2)^2} \tag{2}$$

Also,

$$\lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=0$$
$$\lim_{h\to 0}\frac{f(0,h)-f(0,0)}{h}=0$$

shows $[D_1 f(0,0)]=0$ and $[D_2 f(0,0)]=0$.

$$\begin{aligned}\limsup_{(x,y)\to(0,0)}\left|\frac{x^5-4x^3y^2-xy^4}{(x^2+y^2)^2}\right| &\le \lim_{(x,y)\to(0,0)}|x|\left|\frac{x^4}{(x^2+y^2)^2}\right|+\lim_{(x,y)\to(0,0)}|4x|\left|\frac{x^2y^2}{x^2+y^2)^2}\right|\\ &\quad+\lim_{(x,y)\to(0,0)}|x|\left|\frac{x^4}{(x^2+y^2)^2}\right|\\ &\le \lim_{(x,y)\to(0,0)}|x|+\lim_{(x,y)\to(0,0)}|4x|+\lim_{(x,y)\to(0,0)}|x|=0\end{aligned}$$

shows $D_2 f$ is continuous at $(0,0)$, and similar bounding shows $D_1 f$ is continuous at $(0,0)$ as well.

(b)

(1) and (2) show that away from $(0,0)$, $D_{12}f$ and $D_{21}f$ exist and are continuous. At $(0,0)$,

$$[D_{21}f(0,0)]=\lim_{h\to 0}\frac{\frac{h(-h^4)}{h^4}-0}{h}=-1$$
$$[D_{12}f(0,0)]=\lim_{h\to 0}\frac{\frac{h^5}{h^4}-0}{h}=1$$

Since $[D_{21}f(0,0)]\neq[D_{12}f(0,0)]$, one concludes that neither $D_{21}f$ nor $D_{12}f$ are continuous at $(0,0)$.

(c)

See (b).

Q28

For a fixed $t>0$, $\varphi(x,t)$ is continuous on each piece. Further, the values of each piece match at $x=\sqrt{t}$ and $x=2\sqrt{t}$, so φ is continuous for all x for this fixed $t>0$. By symmetry, φ is continuous at all (x,t) for $t<0$. For points at $t=0$, notice that for every (x,t) such that $t\neq 0$,

$$|\varphi(x,t)|\le\sqrt{|t|}$$

Hence in arbitrary small neighborhood around any point $(x,0)$, the absolute value of φ is arbitarily small. This shows that φ is continuous on all of $\mathbb{R}^2$.

Next, consider

$$\lim_{h\to 0}\frac{\varphi(x,h)-0}{h}\tag{3}$$

By definition of φ it is clear (3) is 0 when $x \leq 0$. When $x > 0$, it is also clear that as h gets arbitrarily small, x lands on the "otherwise" part of φ, hence (3) is 0 once again. This shows $[D_2\varphi(x,0)] = 0$.

Assume $|t| < 1/4$. Then $2\sqrt{t} < 1$, meaning $\varphi(x,t) = 0$ when $x \notin [0, 2\sqrt{t}]$ portion of $[-1,1]$.

$$\begin{aligned}
f(t) &= \int_0^{\sqrt{t}} x dx + \int_{\sqrt{t}}^{2\sqrt{t}} (-x + 2\sqrt{t}) dx \\
&= \frac{1}{2} x^2 \Big|_0^{\sqrt{t}} + \left[-\frac{x^2}{2} + 2\sqrt{t}x \right]_{\sqrt{t}}^{2\sqrt{t}} \\
&= \frac{t}{2} + (-2t + 4t - (-\frac{t}{2} + 2t)) \\
&= t
\end{aligned}$$

So $f(t) = t$ for $t \in [-1/4, 1/4]$. This gives $f'(0) = 1$. However, $[D_2\varphi(x,0)] = 0$ for every x, so

$$\int_{-1}^{1} [D_2\varphi(x,0)] dx = 0.$$

In conclusion,

$$\int_{-1}^{1} [D_2\varphi(x,0)] dx \neq f'(0)$$